

Lecture 1: Diversification and Mean-Variance

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FINM 36700: Portfolio Management

Outline

The Big Picture

Diversification

Mean-Variance

Excess Returns

Appendix



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Key results

- ▶ Portfolio risk is a nonlinear function of security risk.
- ▶ If we assume frictionless markets, then we can analytically solve for “optimal” return-risk allocations.
- ▶ The optimal formula penalizes securities for marginal risk (covariance), not total risk (volatility.)
- ▶ The result is analytical, efficiently implemented, and maximizes portfolio Sharpe Ratio.

You should not care how volatile Bitcoin is
You should care the additional vol of Bitcoin compared to vol of assets you hold right now = marginal risk



Key questions

- ▶ What do we mean by “optimal”?
- ▶ What is the right measure of risk?
- ▶ How do we forecast returns?
- ▶ How well does this work in practice?



Portfolio Management Procedure

1. Define the security universe
2. Model security risk and performance
3. Forecast returns
4. Define the portfolio's objective
5. Define portfolio's constraints
6. Simulate the candidate portfolios
7. Optimize among the portfolios
8. Assess the constructed portfolio's performance



Getting Started

To begin,

- ▶ we need to understand diversification and non-linearity of risk.
- ▶ we will examine the full portfolio optimization process.

During the course, we will dig deeper into each of the steps of the process.



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Return notation: one-period

notation	description	formula	example
r^i	return rate of asset i		
r^f	risk-free return rate		
$\tilde{r}^i$	excess return rate of asset i	$r^i - r^f$	



Two investments: bonds and stocks

Consider the following portfolio example

Table: Portfolio example

	return	allocation weight
bonds	r^b	w
stocks	r^s	$1 - w$

Table: Return statistics notation

mean	variance	correlation
μ	σ^2	ρ



Portfolio return stats

Investment portfolio return r^p has mean and variance of

$$\mu^p = w\mu^b + (1 - w)\mu^s$$

$$\sigma_p^2 = w^2\sigma_b^2 + (1 - w)^2\sigma_s^2 + 2w(1 - w)\rho\sigma_s\sigma_b$$



Perfect correlation

Suppose that $\rho = 1$.

- ▶ Then the volatility (standard deviation) of the portfolio is proportional to the asset allocation weights:

$$\sigma_p = w\sigma_b + (1 - w)\sigma_s$$

- ▶ Thus, both mean and volatility are linear in the allocations.



Imperfect correlation

Suppose that $\rho < 1$.

- ▶ The volatility function is convex,
this is sub additivity
sum of total < sum of components

$$\sigma_p < w\sigma_b + (1 - w)\sigma_s$$

- ▶ Yet the mean return is still linear in the portfolio allocation:

$$\mu^p = w\mu^b + (1 - w)\mu^s$$



Diversification

Portfolio **diversification** refers to this case where

- ▶ mean returns are linear in allocations
- ▶ while volatility of returns is less than linear in allocation.

This only required $\rho < 1$.

if i add an asset in the portfolio that is 80% correlated to another asset, i will still have diversification
As long as correlation is less than 1, there will be sub-additivity and we have some diversification



A perfect hedge

For $\rho = -1$,

- ▶ The portfolio variance can be as small as desired, by choosing the appropriate allocation, w .
- ▶ In fact, $\sigma_p = 0$ if

$$w = \frac{\sigma_s}{\sigma_b + \sigma_s}$$

- ▶ Thus, a riskless portfolio can be formed from the two risky assets.

but your return might be very low (risk free return) because your portfolio is riskless



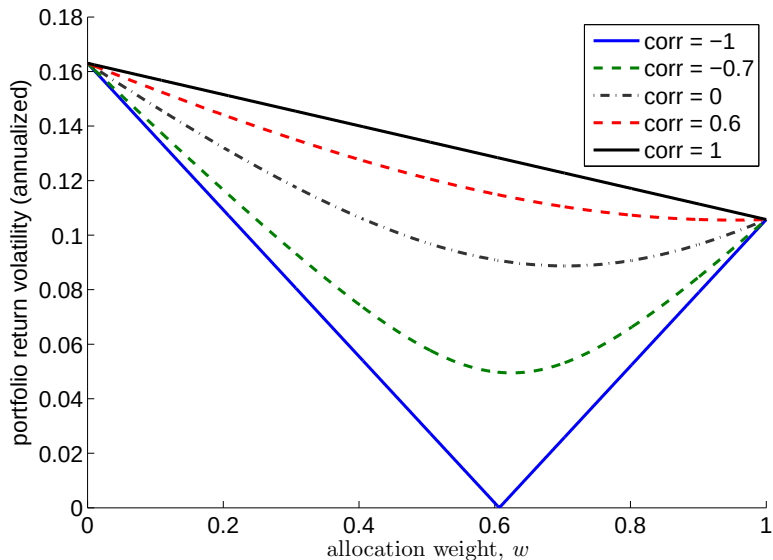


Figure: Diversification of investment portfolio between two risky assets.

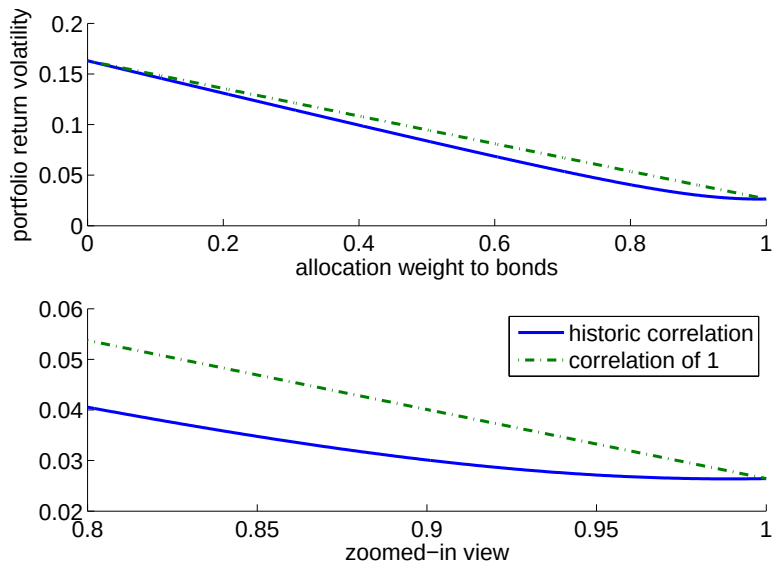


Figure: Diversification over investment in U.S. market index and 10-year T-note. Source: **CRSP** and **N.Y. Fed**. July 1971 to June 2012.

Allocation among n assets

Consider the following portfolio allocation problem:

- ▶ n risky securities,
- ▶ return **volatility (std.dev.)** denoted σ_i
- ▶ return **covariance** between security i and j denoted by $\sigma_{i,j}$.
- ▶ w^i denotes the fraction of the portfolio allocated to asset i , with $\sum_{i=1}^n w^i = 1$.

Then

$$\sigma_p^2 = \sum_{j=1}^n \sum_{i=1}^n w^i w^j \sigma_{i,j}$$



Variance of the equally weighted portfolio

Consider an equally-weighted portfolio, with $w^i = 1/n$ for each asset. Then

$$\sigma_p^2 = \underbrace{\frac{1}{n^2} \sum_{i=1}^n \sigma_i^2}_{\text{sum of variances}} + \underbrace{\frac{1}{n^2} \sum_{j \neq i} \sum_{i=1}^n \sigma_{i,j}}_{\text{sum of covariances}}$$

In the earlier example with bonds and stocks, $n = 2$,

$$\sigma_p^2 = \frac{1}{4} \sigma_b^2 + \frac{1}{4} \sigma_s^2 + \frac{1}{2} \underbrace{\sigma_{b,s}}_{\rho \sigma_b \sigma_s}$$



Portfolio variance as average covariances

Use the following notation for averaging the variances and covariances across the n assets:

$$\text{avg} [\sigma_i^2] \equiv \frac{1}{n} \sum_{i=1}^n \sigma_i^2$$

$$\text{avg} [\sigma_{i,j}] \equiv \frac{1}{n(n-1)} \sum_{j \neq i} \sum_{i=1}^n \sigma_{i,j}$$

So the portfolio variance can be written as

$$\sigma_p^2 = \frac{1}{n} \text{avg} [\sigma_i^2] + \frac{n-1}{n} \text{avg} [\sigma_{i,j}]$$



Portfolio irrelevance of individual security variance

As number of securities in portfolio, n , gets large,

$$\lim_{n \rightarrow \infty} \sigma_p^2 = \text{avg} [\sigma_{i,j}]$$

- ▶ Individual security variance is unimportant!
- ▶ Overall portfolio variance is average of individual security covariance.



Diversified portfolio

Obtained this result using equally-weighted portfolio, $w^i = 1/n$.

- ▶ Don't need equal weighting, just that

$$\lim_{n \rightarrow \infty} w^i = 0$$

- ▶ That is, as n gets large the portfolio must have trivial exposure to security i .
- ▶ This is the sense in which portfolio must be diversified for individual variances to become unimportant.



Portfolio variance decomposition

Above we saw the **equally-weighted** portfolio variance:

$$\sigma_p^2 = \frac{1}{n} \text{avg} [\sigma_i^2] + \frac{n-1}{n} \text{avg} [\sigma_{i,j}]$$

Variance has a term which can be diversified to zero, and another term that remains.

Suppose that asset returns have

- ▶ **identical volatilities**, $\sigma_i = \sigma$
- ▶ **identical correlations**, $\rho_{i,j} = \rho$



Systematic risk

$$\sigma_p^2 = \frac{1}{n}\sigma^2 + \frac{n-1}{n}\rho\sigma^2$$

$$\lim_{n \rightarrow \infty} \sigma_p^2 \rightarrow \underbrace{\rho\sigma^2}_{\text{systematic}}$$

- ▶ A fraction, ρ , of the variance is **systematic**.
- ▶ No amount of diversification¹ can get portfolio variance lower:

$$\sigma_p^2 \geq \rho\sigma^2$$

¹Inequality holds for any n and any set of allocations $\{w^i\}$.



Idiosyncratic risk

$$\sigma_p^2 = \frac{1}{n}\sigma^2 + \frac{n-1}{n}\rho\sigma^2$$

- ▶ **Idiosyncratic** risk refers to the diversifiable part of σ_p^2 .
- ▶ An **equally-weighted** portfolio ² has idiosyncratic risk equal to $\frac{1}{n}\sigma^2$.

²For general weights, w^i , remaining idiosyncratic risk is bounded by $\max_i w^i \sigma^2$.



Correlation and diversified portfolios

$$\sigma_p^2 = \frac{1}{n}\sigma^2 + \frac{n-1}{n}\rho\sigma^2$$

- For $\rho = 1$, there is no possible diversification, regardless of n .

$$\sigma_p^2 = \sigma^2$$

- For $\rho = 0$, there is no systematic risk, only variance is remaining idiosyncratic:

$$\sigma_p^2 = \frac{1}{n}\sigma^2$$

And as n gets large the portfolio is riskless,

$$\lim_{n \rightarrow \infty} \sigma_p^2 = 0$$



Riskless portfolios

- ▶ Above, we found that a riskless portfolio could be created if $\rho = -1$.
- ▶ Here, we found that a riskless portfolio can be created if $\rho = 0$.

Question:

How did the assumptions behind these conclusions differ?



Answer:

- ▶ In the *case of just two underlying assets*, complete diversification is achieved with $\rho = -1$.
- ▶ In the case of many assets, complete diversification is achieved when all assets are uncorrelated, *and the number of assets in the portfolio goes to infinity*.



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Mean-variance comparisons

We want to compare risk and return...

- ▶ Use mean return to score the portfolio's benefits.
- ▶ Use variance (or volatility) of return to score the portfolio's risk.

Consider the case of two assets:



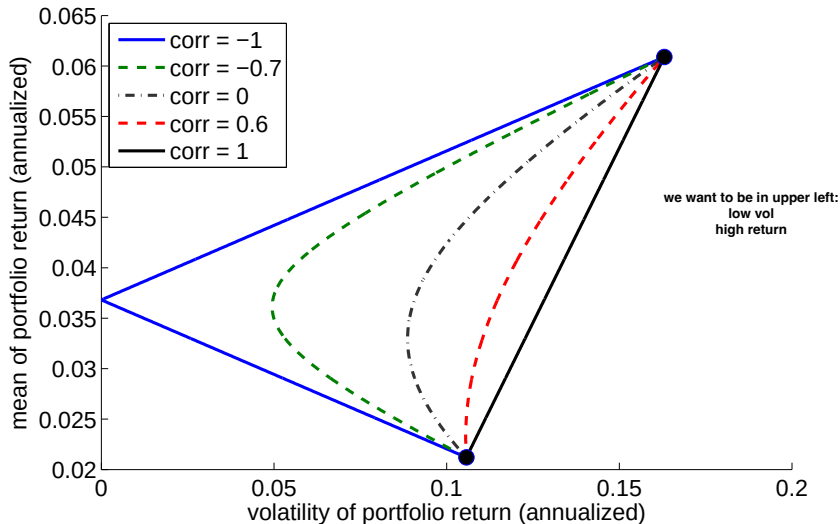


Figure: Example in mean-volatility space of diversification between two assets.

Diversification across n assets

With n securities, there is further potential for diversification.

- ▶ The set of all possible portfolios formed from this basis of assets forms a convex set in mean-variance space.
- ▶ The boundary of this set is known as the mean-variance frontier, and it forms a parabola.
- ▶ The boundary of the set in mean-volatility space forms a hyperbola.

We use **MV frontier** to refer to both the mean-variance and mean-volatility frontiers.



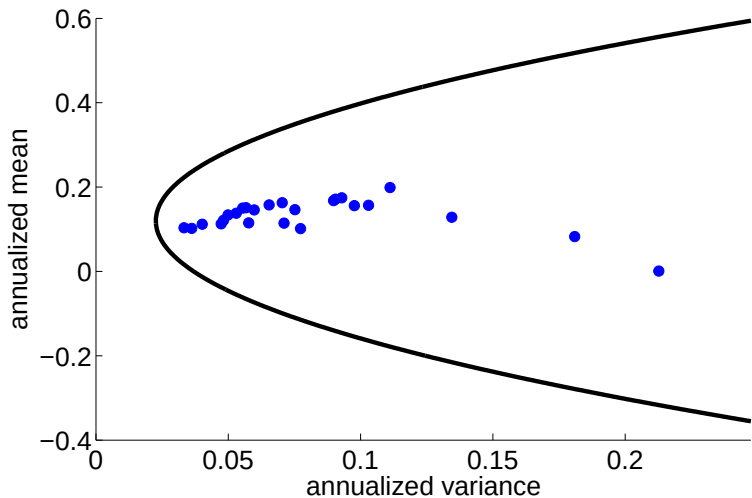


Figure: Mean-variance frontier formed by 25 U.S. equity portfolios, sorted by size and book/market.

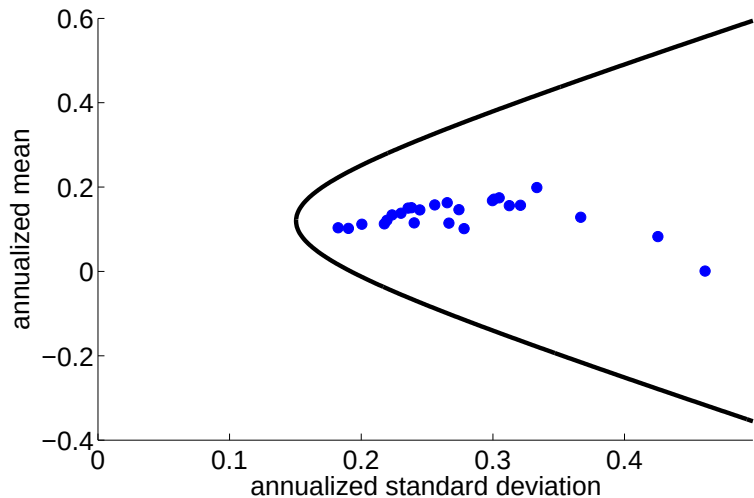


Figure: Mean-volatility frontier formed by 25 U.S. equity portfolios, sorted by size and book/market.

Efficient portfolios

The top segment of the MV frontier is the set of **efficient MV portfolios**.

- ▶ These portfolios **maximize mean return given the return variance**.
- ▶ Contrast this with the lower segment of the MV frontier, the **inefficient MV portfolios**.
- ▶ The inefficient MV portfolios **minimize mean return given the return variance**.



Importance of MV analysis

- ▶ MV analysis is the most widely used tool in portfolio allocation.
- ▶ The model gives a tractable way to balance risk and return.
- ▶ Later in the course, we will a connection between MV analysis and beta-factor models.



Notation

Suppose there are n risky assets.

- ▶ $\mathbf{r}$ is an $n \times 1$ random vector. Each element is the return on one of the n assets.
- ▶ Let $\boldsymbol{\mu}$ denote the $n \times 1$ vector of mean returns. Let $\boldsymbol{\Sigma}$ denote the $n \times n$ covariance matrix of returns.

$$\boldsymbol{\mu} = \mathbb{E}[\mathbf{r}]$$

$$\boldsymbol{\Sigma} = \mathbb{E}[(\mathbf{r} - \boldsymbol{\mu})(\mathbf{r} - \boldsymbol{\mu})']$$

- ▶ For now, we suppose no risk-free rate is available.
- ▶ Assume $\boldsymbol{\Sigma}$ is positive definite—no asset is a linear function of the others.



Portfolios

- ▶ An investor chooses a **portfolio**, defined as a $n \times 1$ vector of allocation weights, ω .
- ▶ These allocation weights must sum to unity:

$$\omega' \mathbf{1} = 1$$

you can have 300% of asset A and 200% of asset B
the sum of the weights must be 1 (100%) but you can short and long

where $\mathbf{1}$ denotes a $n \times 1$ vector of ones.

- ▶ No shorting restriction here: elements of ω can be negative.



Return moments

The portfolio return on some portfolio, ω^P , is

$$r^P = (\omega^P)' \mathbf{r}.$$

The portfolio return moments are

$$\mu^P =: \mathbb{E}[r^P] = (\omega^P)' \boldsymbol{\mu}$$

$$\sigma_P^2 =: \text{var}(r^P) = (\omega^P)' \boldsymbol{\Sigma} \omega^P$$

$$\text{cov}(r^P, \mathbf{r}) = \boldsymbol{\Sigma} \omega^P$$



MV Portfolio

a vector of portfolio weights

A Mean-Variance (MV) portfolio is a vector, ω^* , which solves the following optimization for some number μ^P :

Minimize portfolio variance (risk) $\min_{\omega} \omega' \Sigma \omega$

Given a targeted expected return s.t. $\omega' \mu = \mu^P$

Ensure that the weights sum to 1 $\omega' \mathbf{1} = 1$

- ▶ Note that the objective function is convex in w , given that Σ is positive definite.
- ▶ The constraint set is also convex.
- ▶ Thus, the solution, ω^* is characterized by the first-order conditions.



MV solution

If we are using the excess return, these asset allocation is the one that maximizes the sharpe ratio

Thus, a portfolio ω^* is MV iff exists $\delta \in (-\infty, \infty)$ such that

$$\omega^* = \delta \omega^t + (1 - \delta) \omega^v$$

$$\omega^t \equiv \underbrace{\left(\frac{1}{\mathbf{1}' \Sigma^{-1} \mu} \right)}_{\text{scaling}} \Sigma^{-1} \mu,$$

This is tangency portfolio, has target mean
Tries to maximize sharpe ratio

$$\omega^v \equiv \underbrace{\left(\frac{1}{\mathbf{1}' \Sigma^{-1} \mathbf{1}} \right)}_{\text{scaling}} \Sigma^{-1} \mathbf{1}$$

This is GMV portfolio, no target mean

ω^t and ω^v are themselves MV portfolios ($\delta = 0, 1$)



GMV and zero-tangency portfolios

lowest variance

ω^v is the **Global Minimum Variance (GMV) portfolio**. It solves,

Minimize portfolio variance (risk) $\min_{\omega} \omega' \Sigma \omega$

Ensure that the weights sum to 1 s.t. $\omega' \mathbf{1} = 1$

- This is the same as the MV problem, but dropping the first constraint, ($\omega' \mu = \mu^p$.)

There's no target expected return

ω^t is the portfolio tangent to the mean-volatility frontier and going through the origin. (See next slide.)



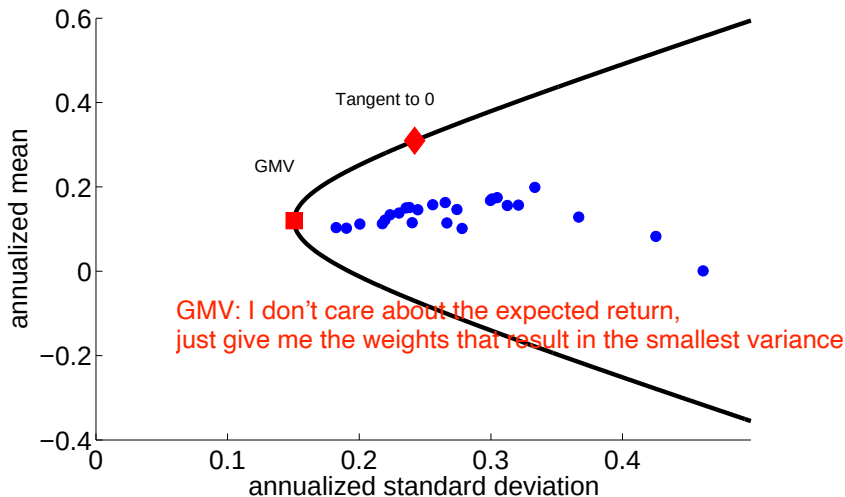


Figure: Illustration of two useful MV portfolios. The Global-Minimum-Variance portfolio as well as the zero-tangency portfolio.

MV investors

Consider **MV investors**, the investors for whom mean and variance of returns are sufficient statistics of the investment.

- ▶ Such investors will hold an MV portfolio, ω^* .
- ▶ Thus, these investors are holding linear combination of just two risky portfolios, ω^t and ω^v .
- ▶ So if in real markets all investors were MV investors, everyone would simply invest in two funds.
- ▶ Those wanting higher mean returns would hold more in the high-return MV, ω^t , while those wanting safer returns would hold more in the low-return MV, ω^v .



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With a riskless asset

Now consider the existence a risk-free asset with return, r^f .

- ▶ Suppose there are still n risky assets available, still notating the risky returns as $\mathbf{r}$
- ▶ Let $\mathbf{w}$ denote a $n \times 1$ vector of portfolio allocations to the n risky assets.
- ▶ Since the total portfolio allocations must add to one, we have

$$\text{allocation to the risk-free rate} = 1 - \mathbf{w}'\mathbf{1}$$



Mean excess returns

μ denotes the vector of mean returns of risky assets, $\mathbb{E}[\mathbf{r}]$.

Let μ^P denote the mean return on a portfolio.

$$\mu^P = (1 - \mathbf{w}'\mathbf{1}) r^f + \mathbf{w}'\mu$$

Use the following notation for excess returns:

$$\tilde{\mu} = \mu - \mathbf{1}r^f$$

Thus the mean return and mean excess return of the portfolio are

$$\mu^P = r^f + \mathbf{w}'\tilde{\mu}$$

$$\tilde{\mu}^P = \mathbf{w}'\tilde{\mu}$$



Variance of returns

- ▶ The risk-free rate has zero variance and zero correlation with any security.
- ▶ Let Σ continue to denote the $n \times n$ covariance matrix of *risky* assets, (and is positive semi-definite.)
- ▶ The return variance of the portfolio, w^P is

$$\sigma_p^2 = w' \Sigma w$$

Covariance matrix is used to quantify the portfolio risk

Enables the optimizer to diversify the risk

Provides both:

Variance of each asset's return (diagonal elements)

How each return move together / covariance (off-diagonal elements)



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The $\tilde{M}\tilde{V}$ problem with a riskless asset

Maximizing return with a given level of risk, or
Minimizing risk with a given level of return

A **Mean-Variance portfolio with risk-free asset** ($\tilde{M}\tilde{V}$) is a vector, $\mathbf{w}^*$, which solves the following optimization for some mean excess return number $\tilde{\mu}^P$:

$$\begin{aligned} \min_{\mathbf{w}} \quad & \mathbf{w}'\Sigma\mathbf{w} \\ \text{s.t.} \quad & \mathbf{w}'\tilde{\boldsymbol{\mu}} = \tilde{\mu}^P \end{aligned}$$

- ▶ In contrast to the MV problem, there is only one constraint.
- ▶ The allocation weight vector, $\mathbf{w}$ need not sum to one, as the remainder is invested in the risk-free rate.



Solving the $\tilde{M}\tilde{V}$ problem

Solving the problem is straitforward:

1. Set up the Lagrangian with just one constraint.
2. The FOC is sufficient given the convexity of the problem.
3. Finally, substitute the Lagrange multiplier using the constraint.

Refer to the solution as an $\tilde{M}\tilde{V}$ portfolio.



$\tilde{M}V$ solution

Because this is a non-linear regression, if the historic mean has an error of 5%, the resulting tangency portfolio will have a significantly higher error (10% or more)
 Here, the output will exacerbate the error problem
 The output will have more noise than the input
 Contradictory to the LLN where the output will have less noise (variability) than input (sample)

$$\mathbf{w}^* = \tilde{\delta} \mathbf{w}^t$$

In this case, we are too dependent to the correlation matrix and historic mean
 If the historic mean is showing that SPY is 50 bps higher than EFA, and the correlation is around 90%
 The tangency portfolio is going to recommend go long 700% SPY and short 600% EFA
 This is to take the 50bps excess return (SPY - EFA)
 But what if EFA suddenly becomes 60 bps higher than EFA on several years? The portfolio will blow up because you're long 700% SPY and short 600% EFA

for the portfolio

$$\mathbf{w}^t = \underbrace{\left(\frac{1}{\mathbf{1}' \Sigma^{-1} \tilde{\mu}} \right)}_{\text{scaling}} \Sigma^{-1} \tilde{\mu}$$

and allocation

$$\tilde{\delta} = \left(\frac{\mathbf{1}' \Sigma^{-1} \tilde{\mu}}{(\tilde{\mu})' \Sigma^{-1} \tilde{\mu}} \right) \tilde{\mu}^P$$



$\tilde{M}\tilde{V}$ portfolio variance formula

The return variance of an $\tilde{M}\tilde{V}$ portfolio is given by

$$\frac{(\tilde{\mu}^P)^2}{(\tilde{\mu})' \Sigma^{-1} \tilde{\mu}}$$

This implies that the return volatility (standard-deviation) is linear in the absolute value of the mean excess return:

$$\frac{|\tilde{\mu}^P|}{\sqrt{(\tilde{\mu})' \Sigma^{-1} \tilde{\mu}}}$$



Tangency portfolio

The result is that any $\tilde{M}\tilde{V}$ portfolio is a combination of the **tangency portfolio**, $\mathbf{w}^t$, and a position in the riskless asset.

- ▶ The tangency portfolio, $\mathbf{w}^t$ invests 100% in risky assets, $\mathbf{1}'\mathbf{w}^t = 1$.
- ▶ $\mathbf{w}^t$ is the unique portfolio which is on the risky MV frontier as well as the $\tilde{M}\tilde{V}$ frontier expanded by the risk-free asset.
- ▶ $\mathbf{w}^t$ is the point on the risky MV frontier at which the tangency line goes through the risk-free rate. (See the figure below.)



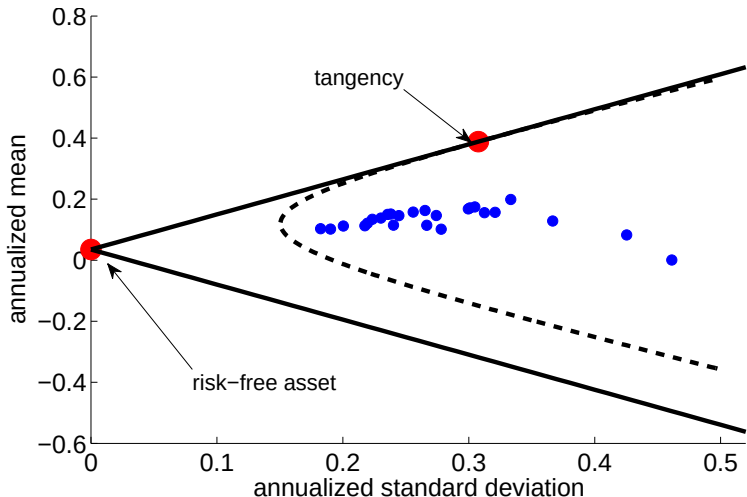


Figure: Illustration of the $\tilde{M}\tilde{V}$ frontier when a riskless asset is available. In this case, the $\tilde{M}\tilde{V}$ portfolio frontier consists of two straight lines. The curved frontier is the MV frontier when a riskless asset is unavailable.

Tangency portfolio and the Sharpe ratio

For an arbitrary portfolio, $\mathbf{w}^P$,

excess return divide by std

$$SR(\mathbf{w}^P) = \frac{\mu^P - r^f}{\sigma^P} = \frac{\tilde{\mu}^P}{\sigma^P}$$

The **tangency portfolio**, $\mathbf{w}^t$, is the portfolio on the risky MV frontier with **maximum** Sharpe ratio.

$$SR(\mathbf{w}^*) = \pm \sqrt{(\tilde{\boldsymbol{\mu}})' \boldsymbol{\Sigma}^{-1} \tilde{\boldsymbol{\mu}}}$$

The SR magnitude is constant across all $\tilde{M}V$ portfolios.
(Sign depends on whether part of the efficient or inefficient frontier.)



Capital Market Line

The **Capital Market Line** (CML) is the efficient portion of the $\tilde{M}\tilde{V}$ frontier.

- ▶ The CML shows the risk-return tradeoff available to $\tilde{M}\tilde{V}$ investors.
- ▶ The slope of the CML is the maximum Sharpe ratio which can be achieved by any portfolio.
- ▶ The inefficient portion of the $\tilde{M}\tilde{V}$ frontier achieves the minimum (negative) Sharpe ratio by shorting the tangency portfolio.



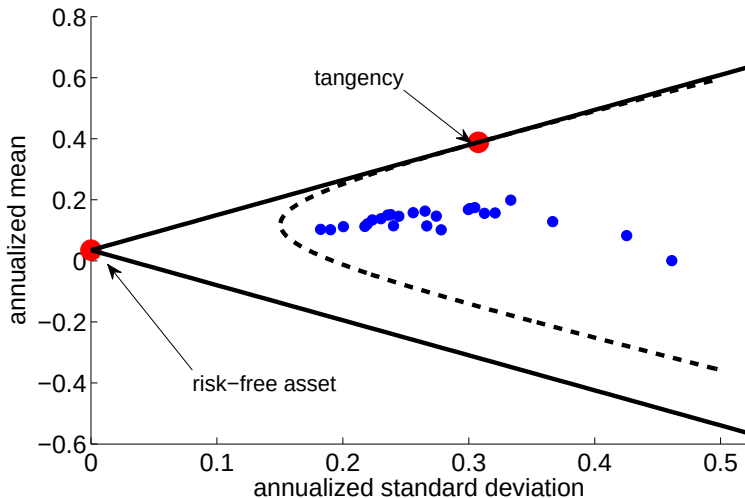


Figure: Illustration of the $\tilde{M}V$ frontier when a riskless asset is available. In this case, the $\tilde{M}V$ portfolio frontier consists of two straight lines. The curved frontier is the MV frontier when a riskless asset is unavailable.

Two-fund separation

Two-fund separation. Every $\tilde{M}V$ portfolio is the combination of the risky portfolio with maximal Sharpe Ratio and the risk-free rate.

Thus, for an $\tilde{M}V$ investor the **asset allocation decision** can be broken into two parts:

1. Find the tangency portfolio of risky assets, $\mathbf{w}^t$.
2. Choose an allocation between the risk-free rate and the tangency portfolio.



Intuition of asset allocation

The two-fund separation says that

- ▶ Any investment in risky assets should be in the tangency portfolio since it offers the maximum Sharpe Ratio.
- ▶ One must decide the desired level of risk in the investment, which determines the split between the riskless asset and the tangency portfolio.



Conclusion

- ▶ Non-additivity of portfolio risk requires us to consider mathematics of diversification.
- ▶ Mean-variance optimization is the dominant approach in industry.
- ▶ But implementation will raise a number of challenges, related to computation and statistics.



References

- ▶ Back, Kerry. *Asset Pricing and Portfolio Choice Theory*. 2010. Chapter 5.

Develops the mathematical formulas for optimization among n assets.

- ▶ Bodie, Kane, and Marcus. *Investments*. 2011. Chapter 7.

Develops the intuition of mean-variance space and optimal portfolios.



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Solving the MV problem: FOC

Solving with Lagrangian multipliers, $(\gamma_1$ and $\gamma_2,$) gives the unconstrained optimization:

$$\mathcal{L} = \frac{1}{2} \omega' \Sigma \omega - \gamma_1 (\omega' \mu - \mu^p) - \gamma_2 (\omega' \mathbf{1} - 1)$$

The first derivative equations are (in matrix notation,)

$$\frac{\partial \mathcal{L}}{\partial \omega'} = \Sigma \omega - \gamma_1 \mu - \gamma_2 \mathbf{1}$$

Get the first-order conditions of optimization by setting equal to zero and solve for ω^* :

$$\omega^* = \Sigma^{-1} \begin{bmatrix} \mu & \mathbf{1} \end{bmatrix} \begin{bmatrix} \gamma_1 \\ \gamma_2 \end{bmatrix}$$



Solving the MV problem: portfolios ω^t and ω^v

Rewrite this as

$$\omega^* = \gamma_1 \Sigma^{-1} \mu + \gamma_2 \Sigma^{-1} \mathbf{1}$$

which can be rewritten as the sum of two portfolios:

$$\omega^* = \gamma_1 (\mathbf{1}' \Sigma^{-1} \mu) \omega^t + \gamma_2 (\mathbf{1}' \Sigma^{-1} \mathbf{1}) \omega^v$$

where

$$\omega^t \equiv \frac{1}{\mathbf{1}' \Sigma^{-1} \mu} \Sigma^{-1} \mu, \quad \omega^v \equiv \frac{1}{\mathbf{1}' \Sigma^{-1} \mathbf{1}} \Sigma^{-1} \mathbf{1}$$



Solving the MV problem: eliminate γ_2

Note that ω^t and ω^v are proper portfolios:

$$(\omega^t)' \mathbf{1} = 1, \quad (\omega^v)' \mathbf{1} = 1$$

Given that $\mathbf{1}'\omega^* = \mathbf{1}'\omega^t = \mathbf{1}'\omega^v = 1$, the equation above implies

$$1 = \gamma_1 (\mathbf{1}'\Sigma^{-1}\mu) + \gamma_2 (\mathbf{1}'\Sigma^{-1}\mathbf{1})$$

Use this to rewrite the MV vector as

$$\omega^* = \delta \omega^t + (1 - \delta) \omega^v$$

where

$$\delta \equiv \gamma_1 (\mathbf{1}'\Sigma^{-1}\mu)$$



MV formulas

For any MV portfolio ω^* , consider the mean, μ^p and variance σ_p^2 ,

Sub out γ_1 to get δ in terms of μ^p ,

$$\delta = \frac{\mu^p - \mu' \omega^v}{\mu' \omega^t - \mu' \omega^v}$$

The return variance, σ_p^2 , is a quadratic function of μ^p ,

$$\sigma_p^2 = \frac{1}{\phi_0 \phi_2 - \phi_1^2} \left[\phi_0 - 2\phi_1 (\mu^p) + \phi_2 (\mu^p)^2 \right]$$

where the coefficients, ϕ are characterized by

$$\phi_0 = \mu' \Sigma^{-1} \mu, \quad \phi_1 = \mu' \Sigma^{-1} \mathbf{1}, \quad \phi_2 = \mathbf{1}' \Sigma^{-1} \mathbf{1}$$

Two-fund separation

Consider any three MV portfolios, ω_a , ω_b , ω^p , which must satisfy the following for some $\delta_a, \delta_b, \delta_p$,

$$\omega_a = \delta_a \omega^t + (1 - \delta_a) \omega^v$$

$$\omega_b = \delta_b \omega^t + (1 - \delta_b) \omega^v$$

$$\omega^p = \delta_p \omega^t + (1 - \delta_p) \omega^v$$

- ▶ ω^t and ω^v are not unique in being able to decompose the MV portfolio, ω^p .
- ▶ Any MV portfolio can be written as a combo of ω_a and ω_b .

$$\omega^p = \vartheta \omega_a + (1 - \vartheta) \omega_b, \quad \vartheta \equiv \frac{\delta_p - \delta_b}{\delta_a - \delta_b}$$



Uncorrelated MV portfolios

Using 2-fund separation, convenient to decompose MV portfolios into two orthogonal portfolios.

- ▶ For any MV portfolio, $\omega^P \neq \omega^v$, there exists another MV portfolio, ω_o such that ω_o orthogonal to ω^P .
- ▶ If ω^P has mean return μ^P , then the orthogonal MV portfolio ω_o has mean return, μ_o , where

$$\mu_o = \frac{\phi_1 \mu^P - \phi_0}{\phi_2 \mu^P - \phi_1}$$

$$\phi_0 = \mu' \Sigma^{-1} \mu,$$

$$\phi_1 = \mu' \Sigma^{-1} \mathbf{1},$$

$$\phi_2 = \mathbf{1}' \Sigma^{-1} \mathbf{1}$$



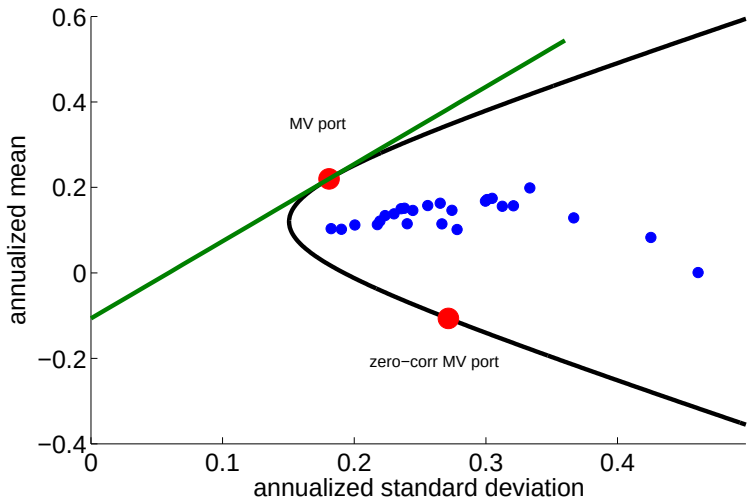
Geometry of uncorrelated portfolios

In mean-volatility space, the orthogonal MV portfolio has a simple geometry.

- ▶ Draw the tangent line at the point of some MV portfolio.
- ▶ Find the value on this tangent line for volatility of zero, (where it hits the vertical axis.)
- ▶ The mean return at this point is, μ_o , the mean return of the orthogonal MV portfolio.

See the following figure.





Figure